Dr. Navneet

(Academic name: Navneet Redhu)

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Professional Summary

Mathematician specializing in applied harmonic analysis and frame theory on LCA groups, with a current research focus on phase retrieval and dynamical sampling for operator-generated frames. Experienced in both pure construction problems (Parseval frames, cyclic frames) and their numerical implementation using MATLAB. Seeking postdoctoral or faculty research positions to extend this work in collaboration with groups working on signal and information processing.

Research Interests

- Applied Harmonic Analysis
- Frame Theory on LCA groups
- Wavelets and Dynamical Sampling
- Phase Retrieval
- Dynamical Frames over Local Fields and Their Applications to Erasures
- Construction of Weaving Information Packets

Experience

National Institute of Science Education and Research (NISER), Bhubaneswar, Odisha, India *2026*

- Postdoctoral Fellow, September 7, 2026 – Present

IILM University, Gurgaon, Haryana, India *2026*

- Assistant Professor, July 1, 2026 – September 1, 2026

Institute of Mathematics for Industry (IMI), Kyushu University, Fukuoka, Japan *2026*

- Visiting Research Fellow, May 5, 2026 – May 18, 2026

Department of Mathematics, Indian Institute of Technology Indore, India *2020–2026*

- CSIR Senior Research Fellow, August 2022 – March 2026
- CSIR Junior Research Fellow, August 2020 – August 2022

Education

Indian Institute of Technology Indore, Simrol, Madhya Pradesh, India *2026*

- Ph.D. in Mathematics (Specialization: Harmonic Analysis, CPI: 8.2 in Coursework).

  Thesis Title: Characterization and Explicit Construction of Pairwise Orthogonal Parseval Frames on LCA Groups (under the supervision of Professor Niraj Kumar Shukla).

  Thesis defended March 10, 2026; degree awarded June 27, 2026.

Indian Institute of Technology Guwahati, India *2019*

- M.Sc. in Mathematics & Computing (CGPA: 6.92/10).

Government P. G. College, Jind, Haryana, India *2016*

- B.Sc. (Pass Course) in Mathematics, Physics, and Statistics (Percentage: 75.41).

Awards and Achievements

- Joint Research Project (Short term Researcher) at IMI, Kyushu University, Japan, May 5 – May 18, 2026
- Institute Travel Grant, IIT Indore for 12th International Conference on Harmonic Analysis & PDEs, Spain, 2025.
- International Travel Grant, ANRF (Govt. of India) for Strobl24 More on Harmonic Analysis, Austria, 2024.

Qualifying Examinations

- Qualified GATE 2020 – **AIR-64**.
- Qualified CSIR-UGC NET (LS) June 2019 – **AIR-21**.
- Qualified CSIR-UGC NET (JRF) December 2019 – **AIR-131**.
- Qualified IIT JAM 2017 – **AIR-179**.

Journal Publications

1. **Navneet Redhu, Anupam Gumber, Niraj K. Shukla** (2025), *Construction of pairwise orthogonal Parseval frames generated by filters on LCA groups*, **Applied and Computational Harmonic Analysis**, **74**, paper No. 101708, 27pp. https://doi.org/10.1016/j.acha.2024.101708

2. **Ole Christensen, Navneet Redhu, Niraj K. Shukla** (2026), *Cyclic frames in finite-dimensional Hilbert spaces*, **Linear Algebra and its Applications**, **728**, 63–81. https://doi.org/10.1016/j.laa.2025.08.016

Submitted Papers

3. **Navneet Redhu, Anupam Gumber, Niraj K. Shukla**, *The geometry of sampling on unions of co-compact subgroups*, **Journal of Mathematical Analysis and Applications**, revision submitted.

4. **Navneet Redhu, Anupam Gumber, Hartmut Führ, Niraj K. Shukla**, *Characterization and construction of pairwise orthogonal Parseval frames with applications to sampling*, **Bulletin des Sciences Mathématiques**, under review. arXiv:2607.13590

Manuscripts in Preparation

5. **Navneet Redhu, Ole Christensen, Niraj K. Shukla**, *Cyclic frames over finite fields.*

6. **Navneet Redhu, Lokendra Kumar, Niraj K. Shukla**, *Dynamical sampling with translation invariant, convolution and Fourier multiplier operators.*

7. **Navneet Redhu, Ryoko Oishi-Tomiyasu**, *Applications of frame theory to diffraction data.*

Invited Talks

- *Wavelet Frame Theory and Shearlets in Data Science and Artificial Intelligence*, Mehta Family School of Data Science and Artificial Intelligence, Indian Institute of Technology (IIT) Roorkee, India (June 15, 2026).
- *An overview of Gabor, wavelet, and shearlet frames: unification and applications*, 83rd La Trobe and Kyushu Joint Seminar on Mathematics for Industry, IMI, Kyushu University, Japan (online, April 28, 2026).

Talks and Poster Presentations

- Talk in 19th International Discussion Meeting in Harmonic Analysis, IISER Mohali, India, DMHA 2025 (December 16–19, 2025).
- Talk in National Conference on Mathematical Analysis and Applications, IIT (ISM) Dhanbad, India, NCMAA 2025 (September 4–6, 2025).
- Poster at 12th International Conference on Harmonic Analysis and Partial Differential Equations, El Escorial, Spain, ICMAT 2025 (June 9–13, 2025).
- Poster at International Conference Strobl24, Strobl, Austria, Strobl24 (June 9–15, 2024).
- Poster at Symposium organized by the Department of Mathematics, IIT Indore, India, IIT Indore (January 11, 2023).

Workshops and Conferences Attended

- Workshop on Harmonic Analysis: Fourier Multipliers and Related Topics, HRI, Prayagraj, India (January 19 – February 2, 2024).
- Conference – 18th International Discussion Meeting in Harmonic Analysis (DMHA), Organized by Department of Mathematics and Computing, IIT Guwahati, India (December 18–21, 2023).
- International Workshop on Representation Theory and Harmonic Analysis, Organized by Department of Mathematics and Computing, IIT Guwahati, India (December 11–16, 2023).
- Online short-term course on Sampling Theory and its Applications: Signal, Image Processing and Data Science, IIT Indore (January 24–29, 2022) at IIT Indore.

Accepted Conference Presentations

- Oral presentation, 3rd International Workshop on Harmonic Analysis and Operator Theory, Institute of Mathematics of the Romanian Academy, Bucharest, Romania, June 9–12, 2026.
- Poster presentation, at International Conference Strobl26, Strobl, Austria, June 2026.

Teaching Experience

- Instructor (Assistant Professor), IILM University, Gurgaon (July – September 2026): Applied Calculus for first-year and Discrete Mathematics for second-year B.Tech students, with lectures and Python laboratory sessions.
- Teaching Assistant, IIT Indore (April 2021 – May 2025, one course in each of nine semesters): Calculus (2021, 2022, 2024); Linear Algebra and ODE-I (2021–2025); Complex Analysis and Differential Equations II; Numerical Methods with MATLAB laboratory (2023). Conducted weekly tutorials for groups of about 50 students, proctored and graded exams, and held additional discussion sessions in my advisor's M.Sc. Topology course.

Service

- Ph.D. Student Representative, IIT Indore (December 2021 – December 2024).
- Postgraduate Senator, IIT Guwahati, during the M.Sc. (April 2018 – March 2019).

Technical Skills

- Programming Languages: C, C++
- Computational Tools: MATLAB, Python, LaTeX

References

Prof. Dr. Niraj Kumar Shukla
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